QueueSmart - Assignment 1

Initial Thoughts and System Design (A1)

https://github.com/COSC-4353-Group-Project

Team Information

Team Member	Primary Responsibility	GitHub Contribution / Commit Notes
Member 1: Justin		
Member 2: Anaya		
Member 3: Vincent		
Member 4: Thu		

1. Initial Thoughts (2 points)

Discuss your initial thoughts on designing QueueSmart. Use the prompts below to draft the group response.

Who are the main users of the system?

Restaurant Customers

How will users interact with the application?

SMS / QR code / web interface

How will administrators interact with the application?

Web Interface, user authenticated

What are the most important features?

Closing Times / Limitation "___ open tables out of ___"

Time Estimate (maybe)

Linear queues (1st in line…50th in line…)

(maybe) pay to win queue system (hey venmo me 5$ ill raise your spot)

Queue cancelations / modifications

What challenges do you anticipate (for example: long queues, notifications, inaccurate wait times, priority handling)?

Queue cleanup (i.e missed appointments / cancelations (have to do a db trigger)

Accurate time models for estimation (totally should put “AI Powered”)

Bypassing the spam wall for sms notifications

User authentication (fake reservations, scalper prevention, Ddos prevention)

2. Development Methodology (2 points)

Which methodology will the team use (Agile / Scrum

Agile

Why is this methodology appropriate for QueueSmart?

Easier to schedule weekly meetings, checkups, corporate speak, yapping jargon

How will the work be divided among the four team members?

Each person takes a **feature** , biased towards strengths and weakness, continue moving forward from there.

How will iterations / reviews work across the later assignments?

Weekly check ins, start from a demo barebones and make sure the foundation is good before you start building upwards. (i.e make it look ugly first, pretty later)

3. High-Level Design / Architecture (6 points)

Required: at least one System Context Diagram. Keep it high-level and simple. Do not include framework, programming-language, or low-level API details.

System Context Diagram Planning

☐ User actor included

☐ Administrator actor included

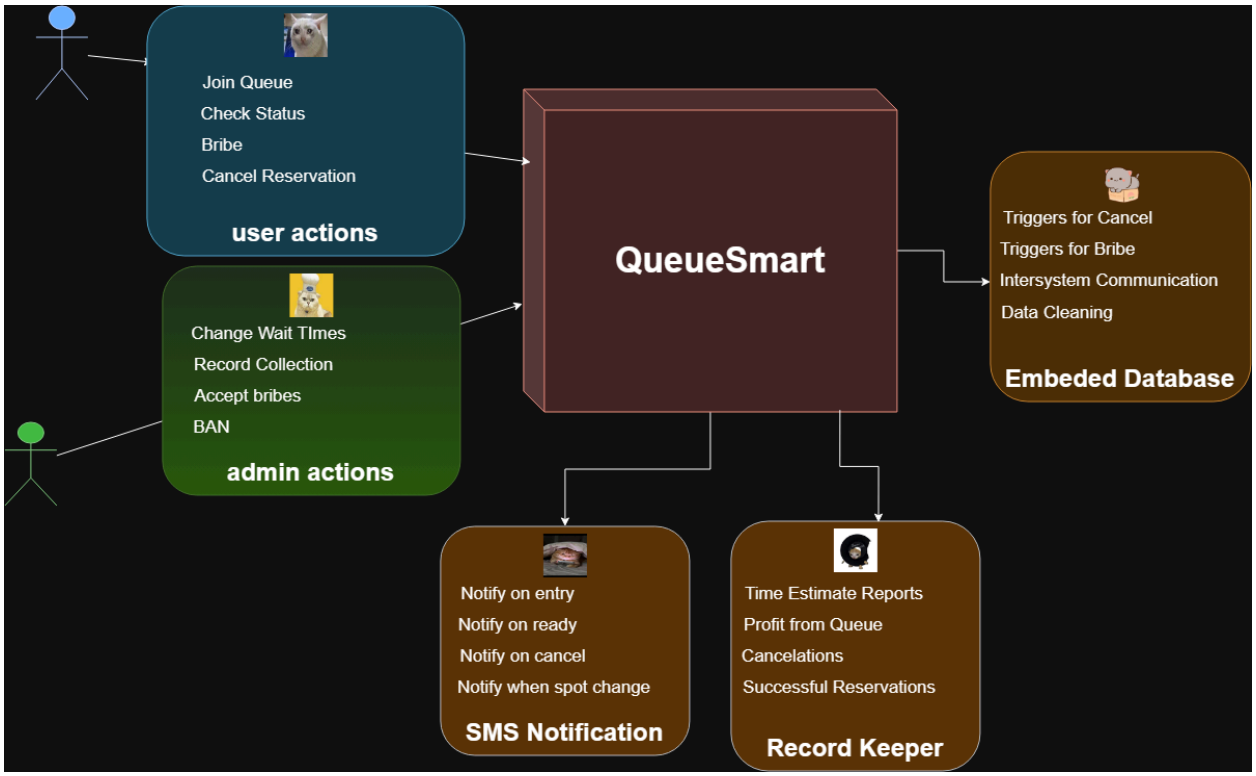
☐ Record Keeper

☐ External notification service

☐ Interaction arrows / labels are clear

☐ Diagram is readable and not overly technical

System Context Diagram below:



Architecture Explanation

Explain how Users interact with QueueSmart.

Users will interact via QRcode forwarded into a website OR online from google maps
Users will be able to Join queue, check status, cancel reservations(via sms maybe), pay to win.

Explain how Administrators interact with QueueSmart.

Admins will be able to login to a secure webpage to manage reservations, generate reports based on reservations, change wait times, and **ban** people (if botting, or poor).

Explain how QueueSmart uses the external notification service.

Queue Smart will use an embedded sms function to send notifications to the user when they register with the QR code (booking details), then will notify customers when their table is ready (or bribed out), and will also be able to be viewed from the website itself.

Briefly explain the overall flow between the major components.

Users will be able to join, cancel, check, and bribe within the web application for QueueSmart. Admins will be able to login securely to manage reservations, change wait times, and even ban people from the QueueSmart system. The QueueSmart system will be able to send out SMS notifications in order to update users of any changes to their queue placement. The system will also be able to keep track of reservations, make cancellations based on time, and figure the profit earned from the bribes. QueueSmart will also have an embedded database in order to have triggers to assist in automating cancellations, bribe adjustments, intersystem communications, and cleaning.

Team Design Decisions

Application type: Web

Development methodology: Agile

Notification approach: ☐ SMS ☐ In-webapp

How queue priority is determined: money

How estimated wait time will be calculated conceptually: based on data from restaurant statistics online, imported into math function that averages party size with wait time. (custom lvm model what if)

Final Submission Checklist

☐ All required questions are answered.

☐ System Context Diagram is included and readable.

☐ Each team member has a meaningful documented contribution.

☐ Each team member has visible GitHub contribution history.

☐ Team contribution table is complete.

☐ Final document is reviewed for consistency and spelling.